

Youngjin Yang

Undergraduate Researcher | Life Sciences

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Research Interest

My interests lie in biological systems engineering and bioprocess development, with an emphasis on designing, evaluating, and optimizing biological systems through careful experimentation and quantitative analysis. During 2026, I worked on RNA engineering research, developing experience in cell-based assays, molecular cloning, reporter-based screening, and experimental reproducibility.

Education

Chung-Ang University 2026 - Present

B.S. in Life Sciences. GPA: 4.20 / 4.50.

Dongguk University 2022.03 - 2024.02

Life Sciences. GPA: 4.09 / 4.50.

Combined GPA: 4.112 / 4.50

Research Experience

Chung-Ang University — Kim Hak-gyun Lab 2026

Research focus: circRNA translation optimization through IRES screening and UTR optimization.

Experimental work: HEK293T cell culture, transient transfection, molecular cloning, and dual-luciferase assays.

Analysis: Quantitative evaluation of IRES candidates and reproducibility across repeated experiments.

Dongguk University — Im Gun-il Lab Jun 2023 - Jan 2024

Cell-based research: Regenerative medicine research, FACS-based evaluation of cellular states in spheroid culture, and bone-marrow preparation.

In vivo research: Participated in mouse and rabbit osteoarthritis-model experiments evaluating therapeutic cell- and drug-related effects, including GFP-based observation.

Selected Research Projects

circRNA Translation Optimization Chung-Ang University · 2026

IRES screening for improving translation efficiency of circular RNA. Firefly/Gaussia dual-luciferase reporter system was used to compare candidates, with FHB confirmed as the WT normalization reference. Follow-up work examined combinations of selected IRES candidates with UTRs and considered repeatability and sequence/structural interactions.

IRES Screening & Reproducibility Analysis 2026

Screened approximately 100 candidate conditions using HEK293T cells. Technical duplicates were measured after 24 h and normalized using Gaussia/Firefly ratios relative to WT (FHB). Repeated screening of the candidate set was used to assess reproducibility and select candidates for downstream optimization.

Technical Skills

Cell & Molecular Biology

HEK293T cell culture · transient transfection · Lipofectamine 3000 · dual-luciferase assay · FACS · PCR · molecular cloning · plasmid preparation

RNA & Biological Systems

circRNA engineering · IRES screening · UTR optimization · reporter assays · experimental design · reproducibility analysis

Data, AI & Tools

Python · quantitative and statistical analysis · RNAfold · feature engineering · AI-assisted coding and debugging · data visualization · interactive mutation-landscape development · GitHub

Leadership & Activities

Student Council *Undergraduate*

Participated in student council activities for two years and served as a department/student council leader.

External & Campus Projects *Undergraduate*

Participated in Korea Social Contribution Association supporter activities and led university projects including curriculum development and creative convergence projects.

Career Direction

Long-term research goal: Improve experimental efficiency through AI-based experimental design and the development of automated laboratory workflows, while enhancing productivity through genetic engineering and control of mammalian cells. I aim to pursue research connecting mammalian cell engineering, quantitative analysis, and biopharmaceutical production.